

Zerui “Jerry” Ma

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RESEARCH INTERESTS

ML Pre/Post-Training & Alignment: Foundation model pre-training (Tiny Recursive Models), Supervised Fine-Tuning (SFT), Parameter-Efficient Fine-Tuning (LoRA, QLoRA), alignment (RLHF, GRPO), multi-GPU Distributed Data Parallel (DDP) scaling on Slurm HPC clusters.

Inference Systems & Low-Precision Acceleration: NVFP4 / FP8 / INT4 quantized GEMM kernel compilation, dynamic precision casting, asynchronous CPU-GPU pinned weight offloading, continuous batching, PagedAttention, and KV cache optimization.

Agentic AI & Generative Architectures: Model Context Protocol (MCP), hierarchical multi-agent coordination, tool calling, GraphRAG with vector/graph databases, and Diffusion Transformers (DiT) for video synthesis and molecular dynamics.

EDUCATION

Southern Methodist University

Dallas, TX

Ph.D. in Electrical Engineering (Moody Graduate Fellow)

August 2026 – Present

Southern Methodist University

Dallas, TX

M.S. in Computer Science (GPA: 3.96 / 4.0)

August 2024 – May 2026

- **Master’s Thesis:** *Predictive Natural Language Metrics of Alzheimer’s Disease and Cognitive Decline Trend Analysis* (analyzing representation geometry, model explainability, and longitudinal cognitive decline).

Southern Methodist University

Dallas, TX

B.S. in Computer Science, B.S. in Data Science, B.A. in Mathematics (GPA: 3.73 / 4.0)

August 2022 – May 2025

RESEARCH & INDUSTRY EXPERIENCE

AT&T

Plano, TX

Research Scientist Intern

June 2026 – August 2026

- Engineered an adaptive semantic LLM cascade router for the Chief Data Office of AT&T, directing user queries to cost-optimal models based on prompt complexity and intent, slashing inference costs by 35% while preserving response quality.
- Pre-trained an edge-capable small language model (SLM) leveraging a Tiny Recursive Model (TRM) architecture with recursive weight-tying, cutting memory footprint by 50% for low-latency on-device inference.
- Investigated out-of-distribution generalization in Supervised Fine-Tuning (SFT), mitigating the knowing-to-doing gap to improve model alignment and instruction-following fidelity on specialized domain tasks.
- Presented research findings and conducted hands-on fine-tuning and evaluation workshops for 250+ enterprise engineers, establishing standardized evaluation benchmarks and parameter-efficient adaptation workflows.

Southern Methodist University

Dallas, TX

Research Assistant

December 2023 – Present

- Post-trained Transformer models (LLaMA, BERT) on long-context clinical narratives with PyTorch, Hugging Face, and Distributed Data Parallel (DDP) across an 8-GPU Slurm cluster, achieving a record 0.70 R^2 regression score.
- Architected an agentic AI inference engine leveraging vLLM for continuous batching and PagedAttention, orchestrating hierarchical planning and tool calling to maximize GPU throughput and reduce latency.
- Deployed containerized local LLM inference services (Docker, Ollama/vLLM) supporting 10,000+ university users with sub-second time-to-first-token (TTFT) and optimized KV cache management.
- Trained conditional diffusion generative models on Slurm HPC clusters to sample continuous molecular dynamics and protein trajectories (submitted to JCTC), bypassing computationally expensive QM/MM calculations.
- Published 6 peer-reviewed papers (AAAI Symposium, NAACL NLP4DH, APA JOPACS, Frontiers in AI, JCTC) advancing ML post-training, NLP representation learning, and agentic systems.

University of Colorado Boulder (Science of Science Lab)

Boulder, CO (Remote)

Research Assistant (Collaborator)

November 2025 – May 2026

- Analyzed large-scale scientific peer-review corpora across top AI/ML venues; developed text processing pipelines using DistilBERT and fine-tuned classifiers to evaluate review depth and detect automated/synthetic text artifacts.
- Investigated longitudinal review quality trends to evaluate the empirical impact of LLMs on scientific peer evaluation.

SELECTED PROJECTS

High-Fidelity AI Video Generation Engine (LTX-2.5) | *PyTorch, CUDA, NVFP4, TensorRT, Diffusers, RTX 5080* 2026

- Implemented an ultra-low-latency video generation pipeline utilizing the LTX-2.5 Diffusion Transformer (DiT) architecture on a consumer NVIDIA RTX 5080 GPU (Blackwell architecture).
- Compiled custom NVFP4 (NVIDIA 4-bit Floating Point) quantized GEMM kernels with dynamic FP4/FP8 casting and robust scale-factor management, reducing model memory footprint by 65% with negligible loss in visual fidelity.
- Engineered an asynchronous, layer-wise pinned-memory weight offloading subsystem that streams DiT weights between system RAM and VRAM during forward passes, enabling multi-frame high-resolution video generation within 16GB VRAM.
- Integrated FlashAttention and CUDA Graph execution across spatio-temporal attention layers, cutting end-to-end inference latency by 42% compared to standard PyTorch implementations.

GPU-Accelerated Quantum Error Detection | *cuQuantum, Qiskit, CUDA, Python* May 2026 – Present

- Simulated quantum circuits with stochastic Pauli error channels using cuQuantum tensor-network statevector backends, accelerating simulation speeds on consumer GPUs.
- Developed a statistical sampling algorithm to pinpoint single- and two-qubit gate errors, reducing syndrome measurement overhead under noisy intermediate-scale quantum (NISQ) regimes.
- Drafting journal publication formalizing the error-mitigation method as a scalable quantum error correction (QEC) verification technique.

Agentic GraphRAG Recommender System | *LangChain, Qdrant, Neo4j, vLLM, FastAPI, Next.js* June 2024 – Present

- **Published at AAAI 2025 Spring Symposium:** Architected an autonomous multi-agent recommendation engine using GraphRAG (Neo4j) and dense vector retrieval (Qdrant) to resolve combinatorial degree requirements.
- Engineered a low-latency inference backend handling 2,000+ concurrent LLM queries with human-in-the-loop (HITL) agent verification, cutting university academic advising turnaround time by 50%.
- Integrated autonomous tool-calling agents with self-hosted vLLM inference endpoints, automating constraint validation and eliminating prerequisite hallucination.

Custom AI Development IDE & Agent Runtime | *FastAPI, Python, TypeScript, Rust, Docker, MCP* May 2024 – Present

- Architected an extensible AI development environment integrating Model Context Protocol (MCP) to orchestrate complex multi-step tool-calling workflows across local and cloud LLMs.
- Built a scalable agent runtime and distributed workflow hub on AWS EC2, enabling persistent memory, dynamic agent delegation, and workflow sharing among developers.
- Engineered an optimized edge inference pipeline for short-context SLMs, minimizing invocation latency and API compute overhead for interactive agent loops.

Conditional Diffusion Models for Protein Trajectories (PyTAO) | *PyTorch, Stable Diffusion, DDP, HPC* 2025 – 2026

- Designed and trained conditional diffusion generative models to sample continuous molecular dynamics and biomolecular trajectories, bypassing expensive quantum mechanical (QM/MM) simulations.
- Built, tested, and released the PyTAO open-source Python library, implementing modular tensor transformations and Slurm parallel training scripts.

Automated Debug Assistant & Concurrency Scheduler OS | *C++, Python, Ollama, Dask, Slurm HPC* Aug 2023 – Dec 2023

- Implemented an automated LLM debugging assistant in C++ interfacing with Ollama REST APIs for iterative diagnostic generation and root-cause analysis in multi-threaded environments.
- Applied mutex locks and deadlock prevention techniques, ensuring stable performance in multithreaded job execution; accelerated computation speeds by 33% using Dask distributed parallelization on HPC systems.

PUBLICATIONS

Language-based AI modeling of personality traits and pathology from life narrative interviews

Oltmanns, J. R., Khandelwal, R., Ma, J., Brickman, J., Do, T., Hussain, R., & Gupta, M. (2025). *Journal of Psychopathology and Clinical Science (APA)*, Impact Factor: 3.9, Citations: 10.

[Google Scholar](#) | [doi:10.1037/abn0001047](https://doi.org/10.1037/abn0001047) | **Contributing Author**

Personality prediction from life stories using language models

Hussain, R., Ma, J., Khandelwal, R., Oltmanns, J., & Gupta, M. (2025). *arXiv preprint arXiv:2506.19258*, Citations: 5.

[Google Scholar](#) | [arXiv:2506.19258](https://arxiv.org/abs/2506.19258) | **2nd Author**

A Recommender System Architecture for University Curriculum Advising

Ma, Z., Hahsler, M., & Moore, P. (2025). *Proceedings of the AAAI Symposium Series*, 5(1), 235–241, Citations: 4.

[Google Scholar](#) | [doi:10.1609/aaais.v5i1.35593](https://doi.org/10.1609/aaais.v5i1.35593) | **1st Author**

Efficient sampling of short protein trajectories with conditional diffusion models

Xiong, C., Kandhan, P., Chen, D., Ma, Z., Smith, E. D., & Tao, P. (2026). *Journal of Chemical Theory and Computation (ACS)*, 22(1), 78, Citations: 3.

[Google Scholar](#) | [doi:10.1021/acs.jctc.5c01089](https://doi.org/10.1021/acs.jctc.5c01089) | **Contributing Author**

Language-based personality assessment from life narratives: a focus on model interpretability and efficiency

Hussain, R., Ma, Z., Khandelwal, R., Oltmanns, J., & Gupta, M. (2026). *Frontiers in Artificial Intelligence*, 9, 1760246, Citations: 2.

[Google Scholar](#) | [doi:10.3389/frai.2026.1760246](https://doi.org/10.3389/frai.2026.1760246) | **2nd Author**

Predictive Natural Language Metrics of Alzheimer’s Disease and Cognitive Decline Trend Analysis

Ma, Z. (2026). *Computer Science and Engineering Theses and Dissertations*, Southern Methodist University.

[Google Scholar](#) | [Master’s Thesis](#)

AI Assistant for Socioeconomic Empowerment Using Federated Learning

Abdelgaber, N., Jahan, L., Castellano, N., Oltmanns, J., Gupta, M., Zhang, J., Pednekar, A., Basavaraju, A., Velazquez, I., & Ma, Z. (2025). *Proceedings of the 5th International Conference on Natural Language Processing for Digital Humanities (NLP4DH at NAACL 2025)*, 490–501.

[Google Scholar](#) | [doi:10.18653/v1/2025.nlp4dh-1.42](#) | [Contributing Author](#)

Scalable Quantum Error Detection via Tensor-Network Statevector Sampling

Ma, Z., et al. (2026). *In Preparation for Journal Submission*. | [Primary Author](#)

CONFERENCES & INVITED TALKS

AAAI 2025 Spring Symposium (Human-Compatible AI Symposia)

San Francisco, CA

Speaker (30-minute Talk): “A Recommender System Architecture for University Curriculum Advising”

March 2025

National Conference for Undergraduate Research (NCUR 2025)

Pittsburgh, PA

Presenter: Poster presentation on “Recommender Systems for University Curriculum Advising”

April 2025

SMU Mayer Research Presentation

Dallas, TX

Speaker (30-minute Talk): “GraphRAG and Agentic Optimization for Student Curriculum Planning”

May 2025

AT&T Enterprise AI Technical Seminar

Plano, TX

Invited Speaker: Technical seminar on “LLM Fine-Tuning, LoRA, and Alignment Workflows” (250+ Attendees)

August 2026

SMU Fall Research Symposium

Dallas, TX

Speaker (15-minute Talk): “Automated Degree Planning with Intelligent Agent Infrastructures”

October 2024

TEACHING & MENTORSHIP EXPERIENCE

Southern Methodist University

Dallas, TX

Teaching Assistant (Machine Learning, Advanced Python, Digital Logic)

January 2025 – May 2026

- Instructed 100+ undergraduate and graduate students in Machine Learning theory, gradient descent, and transformer attention mechanisms; conducted weekly coding labs and office hours.
- Mentored students in implementing ML algorithms, backpropagation, and deep learning architectures from scratch in Python, PyTorch, and scikit-learn.
- Designed laboratory curriculum in digital systems and VLSI testing, guiding students in implementing JTAG boundary-scan controllers and debugging Verilog HDL on Xilinx FPGAs.

Governor’s Champion Summer Camp

Dallas, TX

AI / ML Course Lecturer

July 2024

- Designed and delivered an intensive 4-week Machine Learning course for high-achieving high school students, covering neural networks, intelligent agents, search algorithms, and data preprocessing.

HONORS & FELLOWSHIPS

Moody Graduate Fellowship | *Southern Methodist University*

August 2026

Accelerated Pathways Leadership and Service Scholarship | *SMU*

August 2025

Robert Mayer Undergraduate Research Fellowship | *SMU*

June 2024

Carrie and Edwin Mouzon Mathematics Scholarship | *SMU Department of Mathematics*

August 2023

SMU Discovery Scholarship & Distinguished Scholarship | *SMU*

August 2022

LEADERSHIP & ACTIVITIES

Artificial Intelligence Club (SMU)

Dallas, TX

President

January 2025 – Present

- Lead weekly technical workshops and paper review groups deconstructing complex neural network architectures, including Linear Attention, FlashAttention, Stable Diffusion, RoPE, and GRPO.
- Organize hands-on ML hackathons and guide peers in PyTorch implementation and distributed GPU training workflows.

IEEE-Eta Kappa Nu (IEEE-HKN) Honor Society, Delta Omicron Chapter

Dallas, TX

Recording Secretary

August 2025 – Present

- Elected executive officer managing official records, chapter correspondence, and technical seminar coordination for the premier electrical and computer engineering honor society at SMU.

SMU Rugby Club

Dallas, TX

Winger

August 2023 – Present

- Compete as starting winger representing Southern Methodist University in collegiate rugby union matches and regional conference tournaments.

TECHNICAL SKILLS

ML Training & Alignment (Pre/Post-Training): PyTorch, Hugging Face (Transformers, PEFT, TRL, Datasets), Distributed Data Parallel (DDP), DeepSpeed, Slurm HPC Cluster, Weights & Biases (WandB), Supervised Fine-Tuning (SFT), LoRA / QLoRA, RLHF / GRPO, FlashAttention, Diffusion Transformers (DiT), Conditional Diffusion Models, Tiny Recursive Models (TRM)

Inference Systems & Low-Precision Acceleration: vLLM (Continuous Batching, PagedAttention), TensorRT-LLM, Ollama, NVFP4 / FP8 / INT4 Quantization, Custom CUDA Kernels, Pinned Memory Weight Offloading, KV Cache Management, Semantic Cascade Routing, Edge Inference

Agentic AI Stack & Retrieval: Model Context Protocol (MCP), LangChain, Multi-Agent Systems, Hierarchical Task Planning, Tool Calling / Function Calling, GraphRAG, Vector Databases (Qdrant), Knowledge Graphs (Neo4j), Human-in-the-Loop (HITL)

Languages & Low-Level Computing: Python, C/C++, CUDA C++, Rust, TypeScript, SQL (PostgreSQL), Verilog HDL, Bash/Linux CLI (Ubuntu, Arch), Assembly (x86)

Infrastructure, Cloud & Systems: Docker, Git/GitHub, AWS (EC2, S3), Google Cloud Platform (GCP), Slurm Workload Manager, Linux Kernel & Sysadmin, cuQuantum, Qiskit, FastAPI